

MULTI-TASK RESIDUAL CNN FOR CROSS-DOMAIN MOSQUITO SPECIES CLASSIFICATION

CD-MSC 2026 Challenge Technical Report

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ABSTRACT

Mosquitoes are major vectors of multiple infectious diseases. Acoustic recognition of mosquito flight sounds is a low-cost and non-invasive method for population monitoring and species identification. This technical report presents a Multi-Task Residual Convolutional Neural Network (MTRCNN) developed for the BioDCASE 2026 CD-MSC Challenge, which focuses on cross-domain mosquito species classification. The model adopts Log-Mel spectrogram as input features, and utilizes residual network modules to extract robust acoustic features. Combined with reasonable training strategies and early stopping rules, the system achieves competitive performance on the official test set. Experimental results show that the model performs well on seen classes, while the generalization ability for unseen domains and species still has room for improvement. Future work will focus on domain adaptation and data augmentation to further enhance the cross-domain robustness of the model.

Index Terms — Mosquito classification, Acoustic signal, Residual CNN, Cross-domain generalization, BioDCASE

1. Introduction

Mosquitoes are vectors for numerous infectious diseases, and identifying their species through acoustic signals is a non-invasive and effective monitoring method. This work presents a multi-task residual convolutional neural network (MTRCNN) for mosquito species classification, developed for the BioDCASE 2026 CD-

MSC Challenge. The system aims to improve generalization across unseen data domains while maintaining high accuracy on known species.

2. SYSTEM OVERVIEW

2.1 Feature Extraction

Raw audio recordings are converted into Log-Mel spectrograms, which capture the frequency and temporal patterns of mosquito flight tones. These spectrograms are used as input to the model.

2.2 Model Architecture

The core model is a Multi-Task Residual CNN (MTRCNN). It uses stacked residual blocks to extract hierarchical features from the spectrograms, with the goal of learning both species-specific patterns and domain-invariant representations.

2.3 Training Configuration

- Optimizer: AdamW
- Loss Function: Cross-Entropy
- Hyperparameters:
 - Batch size: 64
 - Maximum epochs: 120
 - Early stopping patience: 10
 - Random seed: 2026

3. EXPERIMENTAL RESULTS

3.1 Overall Performance

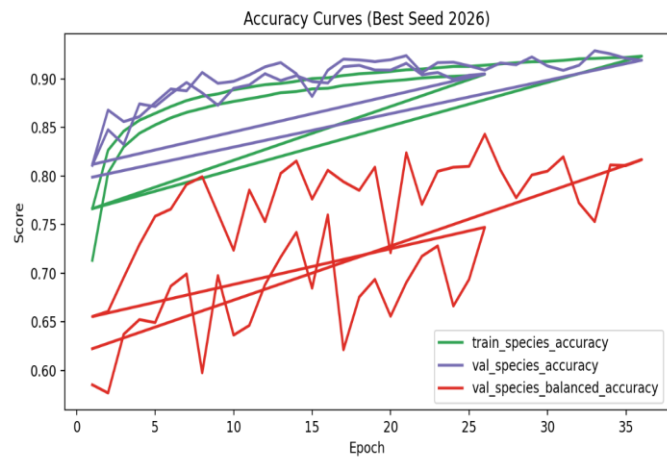
The model's key metrics on the official test set are as follows:

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Metric	Value
Test Accuracy	0.698
Test Species Balanced Accuracy	0.468
Balanced Accuracy (Seen Classes)	0.770
Balanced Accuracy (Unseen Classes)	0.184
Domain Generalization Score	0.586

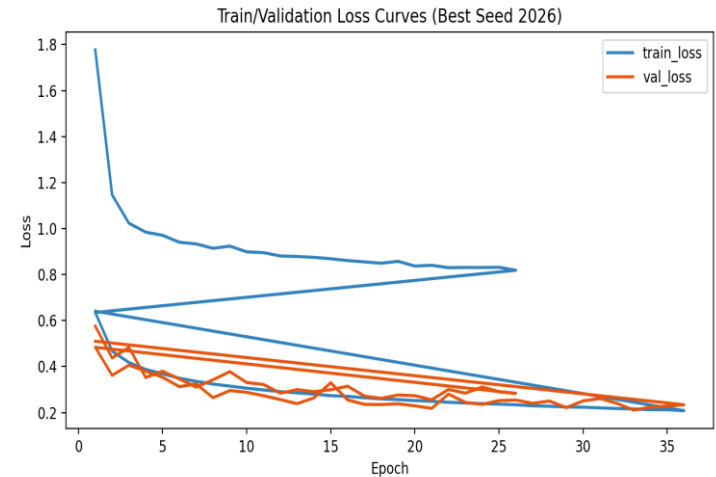
3.2 Training Process

The training curves demonstrate stable convergence and effective generalization on the validation set.



accuracy_curves_best_seed.png

Figure 1: Training and validation accuracy curves across epochs.

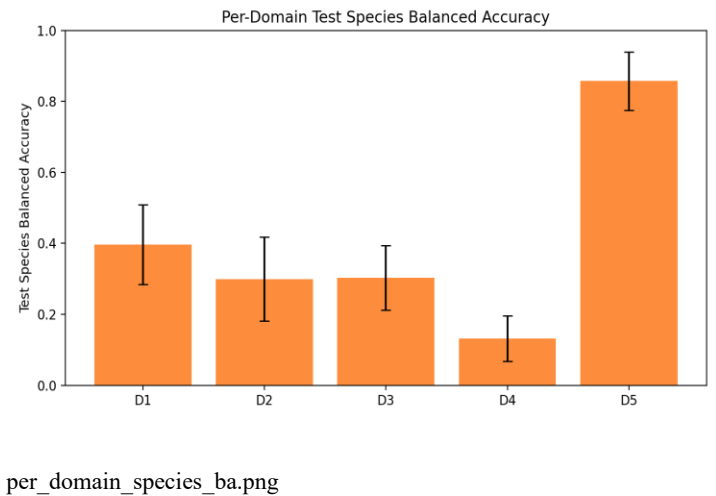


loss_curves_best_seed.png

Figure 2: Cross-entropy loss curves for training and validation.

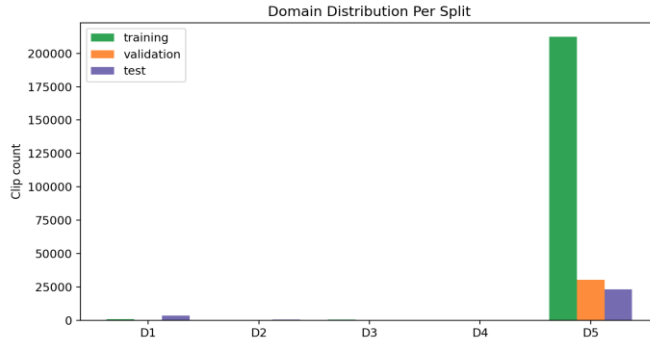
3.3 Species and Domain Analysis

The following visualizations provide insights into the model's performance across different species and domains.



per_domain_species_ba.png

Figure 3: Per-species balanced accuracy across different data domains.



domain_distribution.png

Figure 4: Distribution of training data across different domains.

4. Conclusion

This report introduces an MTRCNN-based system for cross-domain mosquito species classification in the BioDCASE 2026 CD-MSD Challenge. The proposed system achieves good performance on seen species and domains. However, the generalization ability for unseen samples is limited.

In future research, we will introduce advanced domain adaptation algorithms, audio data augmentation and regularization methods to reduce domain offset. We will also optimize the network structure to improve the feature extraction ability for faint acoustic signals, so as to boost the overall performance on cross-domain mosquito classification.

ACKNOWLEDGMENT

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